

4.1 Forces: Turning Effects & Equilibrium

Question Paper

Course	CIE A Level Physics
Section	4. Forces, Density & Pressure
Topic	4.1 Forces: Turning Effects & Equilibrium
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What is the centre of gravity of an object?

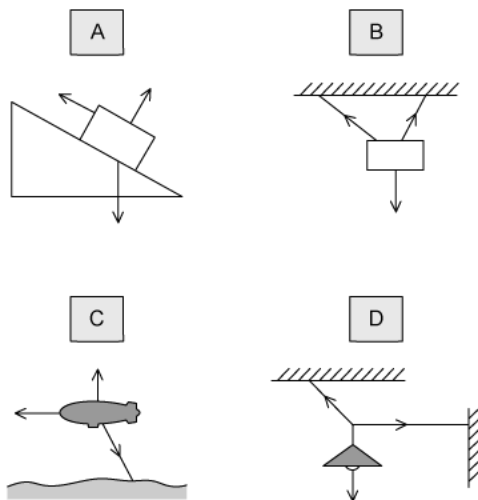
- A. the geometrical centre of the object
- B. the point about which the total torque is zero
- C. the point at which the weight of the object may be considered to act
- D. the point through which gravity acts

[1 mark]

Question 2

The diagrams show the forces acting on different bodies.

Which body **cannot** be in equilibrium?



[1 mark]

Question 3

A force of 5.0 N pushes a ball due north and another force of 12.0 N pushes it due east.

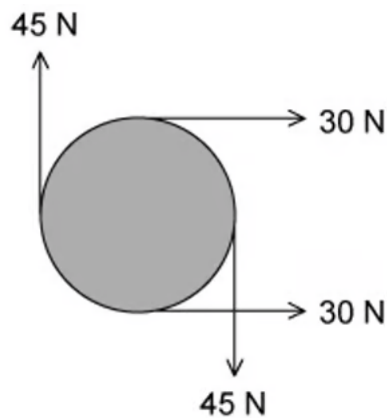
What is the magnitude of the net force acting on the ball?

- A. 7.0 N
- B. 10.9 N
- C. 13.0 N
- D. 17.0 N

[1 mark]

Question 4

The diagram shows four forces applied to a circular object.



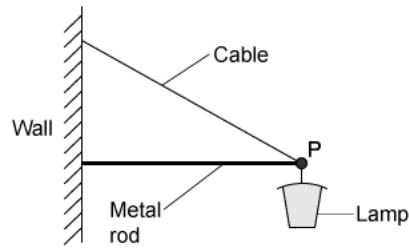
Which of the following describes the resultant force and resultant torque on the object?

	Resultant force	Resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

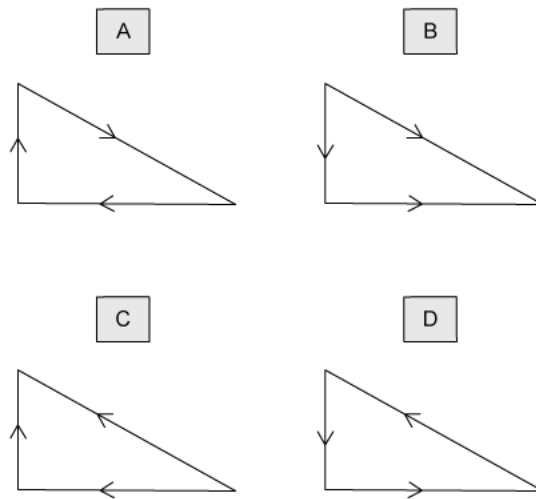
[1 mark]

Question 5

A street lamp is fixed to a wall by a metal rod and a cable.



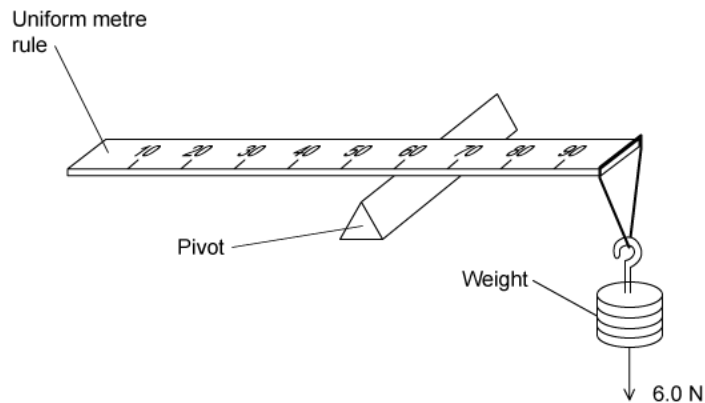
Which vector triangle represents the forces acting at point P?



[1 mark]

Question 6

A uniform metre rule of weight 4.0 N is pivoted at the 60 cm mark. A 6.0 N weight is suspended from one end, causing the rule to rotate about the pivot.



At the instant when the rule is horizontal, what is the resultant turning moment about the pivot?

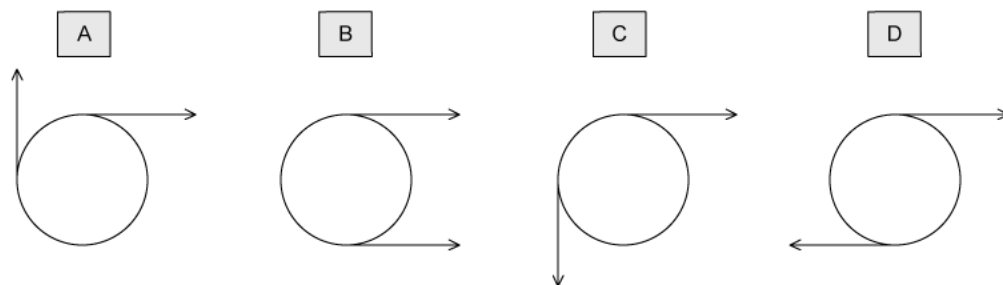
	Resultant moment	Direction
A	2.0 N m	clockwise
B	2.0 N m	anticlockwise
C	2.8 N m	clockwise
D	2.8 N m	anticlockwise

[1 mark]

Question 7

Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude.

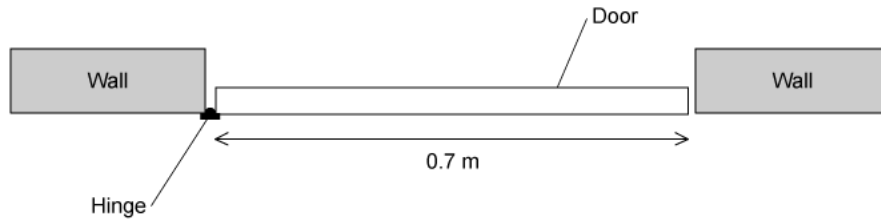
Which arrangement of forces provides only a couple?



[1 mark]

Question 8

The diagram shows a plan view of a door which requires a moment of 14 N m to open it.



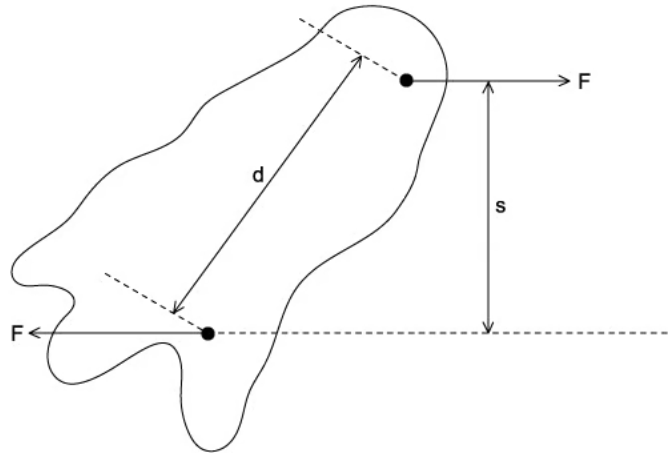
What is the minimum force that must be applied at the door's midpoint to ensure it opens?

- A. 4.9 N
- B. 9.8 N
- C. 20 N
- D. 40 N

[1 mark]

Question 9

Two parallel forces, each of magnitude F , act on a body as shown



What is the magnitude of the torque on the body produced by this couple?

- A. Fd
- B. Fs
- C. $2Fd$
- D. $2Fs$

[1 mark]

Question 10

Forces of 3 N, 4 N and 5 N act at one point on an object. The angles at which the forces act can vary.

What is the value of the minimum resultant force of these forces?

- A. Zero
- B. between 0 and 2 N
- C. 2 N
- D. between 2 N and 4 N

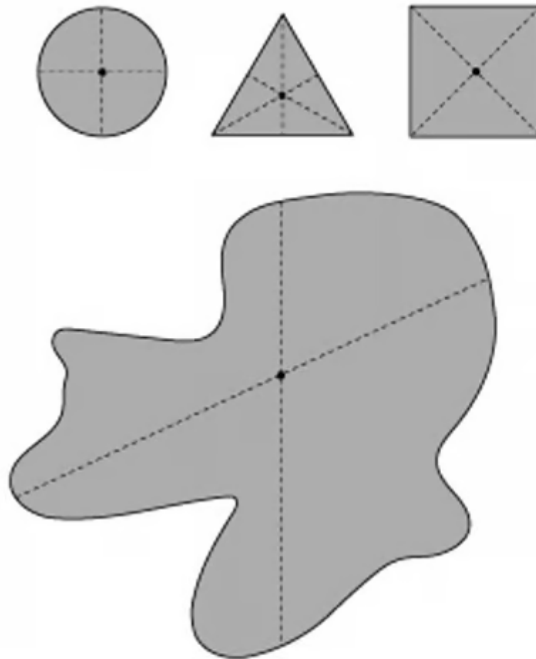
[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- The centre of gravity of an object is defined as the point where all the weight of the object may be considered to act
- Here are some examples of the centre of gravity of some regular and irregular shapes :



A is incorrect because this is true for regular shapes but not for irregular shapes

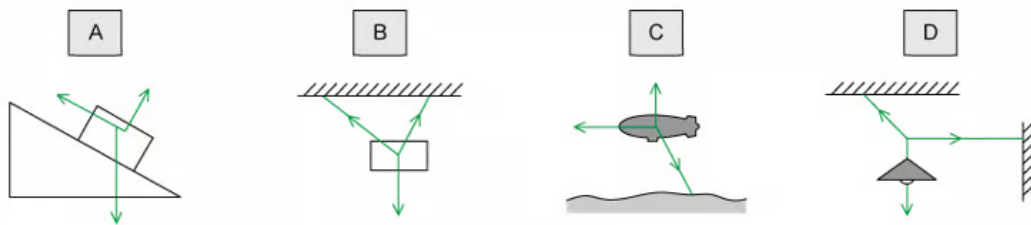
B is incorrect because the point about which the total torque is zero is the point the object is in rotational equilibrium

D is incorrect because the centre of gravity is where the **force of gravity** (weight) acts throughout the object

2

The correct answer is **A** because:

- A system of forces will be in equilibrium if the vector arrows form a closed triangle. This approach is not going to work for this question as it can be argued that the systems could be in equilibrium if the magnitudes of the forces were changed
- However, the question refers to “bodies” rather than “systems”
- Another way to look at it is that three forces can only be in equilibrium if they all pass through the same point, otherwise there will be a turning force on the object
- So, if we join the vector arrows to see where they meet, we can see that the only option where the three forces do not meet at a singular point is option A

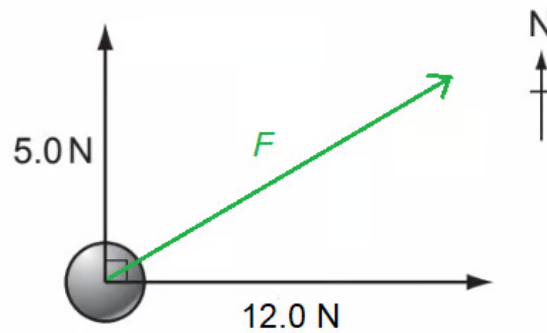


3

The correct answer is **C** because:

- To find the net force for two forces which are perpendicular to each other we can use

$$F = \sqrt{F_N^2 + F_E^2} = \sqrt{5^2 + 12^2} = 13.0$$



4

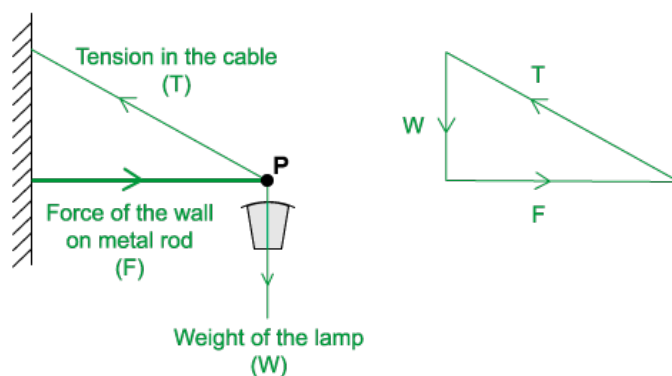
The correct answer is **A** because:

- Vertical resultant force = $45 - 45 = 0$
- Horizontal resultant force = $30 + 30 = 60 \text{ N}$ to the right
- Therefore, the resultant force is non-zero
- The two 45 N forces along with the upper 30 N force all cause a clockwise moment, although the lower 30 N causes an anticlockwise moment, overall, the resultant torque is non-zero

5

The correct answer is **D** because:

- Weight always acts downwards
- Tension is the pull of the cable on the object so it always acts away from the mass
- If we label the forces on the diagram, we can determine the direction of each force and construct a triangle as follows :



A & C are incorrect because weight always acts downwards but these triangles show the weight acting upwards

B is incorrect because the tension in a wire always acts away from the mass whereas this triangle shows it acting towards the mass

6

The correct answer is **A** because:

- To calculate moments, we will need to use
 - $\text{Moment} = \text{Force} \times \text{perpendicular distance from pivot}$
- We will look at the clockwise and anticlockwise moments separately
- The metre rule is uniform, meaning its weight acts at the centre mark, 50 cm
- Therefore, the distance between the pivot and the centre of gravity is 10 cm
- The weight of the ruler produces an anticlockwise moment of
 - $4.0 \times (-0.1) = -0.4 \text{ N m}$
- The 6.0 N weight acts downwards at a distance of 40 cm from the pivot, producing a clockwise moment of
 - $6.0 \times 0.4 = 2.4 \text{ N m}$
- Resultant turning moment = sum of clockwise and anticlockwise moments
 - $2.4 - 0.4 = 2.0 \text{ N m}$
- Therefore, the resultant turning moment is 2.0 N m clockwise

B & D are incorrect because a positive value is generally taken as a clockwise moment

C is incorrect because this is the answer you get if you add the moments together instead of taking away the anticlockwise moment

7

The correct answer is **D** because:

- The system of forces must provide a resultant torque, but we just need a couple so the resultant force should be zero as the forces should cancel out
- To form a couple, two forces must be equal in magnitude, parallel, but opposite in direction, and separated by a perpendicular distance
- Option D is the only one to satisfy this definition

A, B & C are incorrect because the directions of each of these force pairs do not oppose each other

8

The correct answer is **D** because:

- *Moment = Force $\times$ perpendicular distance from pivot*
- The midpoint of the door is at 0.35 m
- So minimum force = moment / distance from hinge (pivot) to midpoint
 - $\frac{14}{0.35} = 40 \text{ N}$

9

The correct answer is **B** because:

- A couple refers to two parallel forces that are equal in magnitude, parallel but opposite in direction and are at a **perpendicular** distance from each other
- The perpendicular distance is s
- So, the torque exerted by the couple is Fs

A is incorrect because the distance d is the distance between the forces, not the perpendicular distance

C & D are incorrect because the moment of a couple is calculated by multiplying the size of **one** of the forces by the perpendicular distance

10

The correct answer is **A** because:

- The minimum resultant force is found by finding the combination of forces which produces the smallest value
- Your first instinct might be to put the two smaller 3 N and 4 N forces together and take away the larger 5 N to make 2 N but there is a combination which can make zero
- Consider the resultant magnitude of 3 N and 4 N when they are perpendicular to each other : $\sqrt{3^2 + 4^2} = 5 \text{ N}$
- So, if the 5 N force is then placed in the opposite direction to the resultant force of the perpendicular 3N and 4N forces, the overall force of the system is zero

